ESC204 | Praxis III

Design Dossier Submission #2 Handout

Due Date: Monday March 30, 2026

Assignment Deadline:	Monday March 30, 2026 | 11:59AM
Team or Individual:	Teams
Assignment Weight:	28%

For this assignment, you will continue to work in your project team to to design, build and test a Prototype System which explores and/or verifies some function(s) of the Design Concept that you developed in Phase 2. You will again create a Design Dossier deliverable containing design and teamwork artifacts to support your process and outcomes: the assignment will be assessed based on this Dossier. This handout provides more information about the assignment, as well as the deliverables you should produce. As you read it, you should refer to the following additional materials:

- The Design Dossier you produced for Submission #1, which contains information about the Design Concept your team is developing a Prototype System to explore;
- The Design Dossier Submission Point #2 Template, posted on Quercus; and
- The Design Dossier Submission Point #2 Rubric, as well as the rubric for the Visual Abstract.

1 Overview of Design Dossier Submission #2

The goal for the remainder of Praxis III is to create a Prototype System that provides insights into how the Design Concept that you proposed in Phase 2 functions. You will design a Prototype System based on your Design Concept, fabricate its subsystems using light fabrication techniques you've practised throughout the course, then integrate the subsystems to form a complete, functional Prototype System. Finally, you'll test your Prototype System against a Specification you've developed to verify its functionality and performance.

The purpose of submitting a Design Dossier at Submission #2 is fourfold:

1. To provide evidence of how your team has investigated at least one **key function** of your final Design Concept through the design, build and verification of a **Prototype System**.
2. To discuss the implementation your Prototype System and the extent to which it meets (or fails to meet) the **Specification** you have developed.
3. To highlight how the performance of your Prototype System relative to the Specification helps to clarify performance of the Design Concept against the overall Engineering Requirements.
4. To present and reflect on both the **team process** and **design process** undertaken during Phase 3.

In addition to Design Dossier Submission #2, your team will also participate in a **Design Debrief** in Studio 13A. Information about the Design Debrief will be provided in another handout. At the

Design Debrief, you will present information from your Phase 3 Design Dossier to the teaching team, as well as physically demonstrating your complete Prototype System.

Table 1 provides an overview of the assignment timeline.

Table 1: Assignment timeline. Deadlines are **bolded**.

Date	Event	Notes
Feb. 18	DD2 materials released	
Feb. 23–27	Prototype System Design	
Feb. 26–27	Soldering Workshop (Optional)	Studio 08B
Mar. 02–17	Subsystem Implementation	
Mar. 17/18	**Subsystem Check-In**	Subsystem fabrication complete
Mar. 22	**Large Order Deadline**	Last day for large DigiFab orders
Mar. 19-30	System Integration & Verification	
Mar. 30	**Design Dossier Submission # 2**	Team Submission

1.1 Submission Checklist

You should submit the following artifacts as part of Design Dossier Submission #2, in advance of the deadline on **March 30, 2026**:

- ☐ A .pdf of your Visual Abstract, as described in Section 4.1. you should submit this to the Design Dossier Submission #2 – Visual Abstract Assignment on Quercus.
- ☐ A Design Dossier, as described in Section 4.2. To submit, you should upload your Design Dossier to SharePoint (or optionally, you may upload a link to your your Design Dossier housed outside of SharePoint, along with a .zip file with identical contents). Your Design Dossier should contain the following Required Artifacts:
  - ☐ Specific Outcome Artifacts re-submitted from Design Dossier Submission #1, with updates made as necessary (Section 4.2.1).
  - ☐ A Specification for your Prototype System (Section 4.2.2).
  - ☐ A Prototype Overview Document (Section 4.2.3).
  - ☐ A completed Verification Protocol & Results spreadsheet (Section 4.2.4).
  - ☐ Process Artifacts supporting the design of your Prototype System (Section 4.2.5).
  - ☐ Process Artifacts supporting the fabrication of the subsystems of your Prototype System (Section 4.2.6).
  - ☐ Process Artifacts supporting the process of integrating the subsystems you fabricated into your completed Prototype System (Section 4.2.7).
  - ☐ Process Artifacts generated during verification of your Prototype System (Section 4.2.8).
  - ☐ Specific Process and Outcome Artifacts which illustrate the Team Process throughout Phase 3 (Section 4.2.9).

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2 Completing the Assignment

You should approach this project by **completing the following steps**:

1. **Carefully read** the materials provided including:
   - this handout;
   - the associated rubric; and
   - the Design Dossier Template for Design Dossier Submission #2.
2. **Establish** what **main design and fabrication tasks** you must complete, paying careful attention to deadlines.
3. Complete, through **teamwork**, **design and fabrication** of your Prototype System, while **creating the required artifacts** for the Design Dossier submission as you go.
4. **Review and organize** the resulting Design Dossier, ensuring you've included all required Process Artifacts.
5. **Create required Outcome Artifacts** based on your collected process artifacts[1].

PAUSE: Note that the Submission Point #2 Handout, Rubric, and Design Dossier Template are meant to be considered together: please ensure you have read or reviewed **all of these** before getting too far into the project! The following sections provide more information about how to use each element as you complete the assignment.

2.1 Using the Handout

Assignment Handouts in Praxis III (like this one!) provide the following key information:

1. How to **approach** the assignment.
2. What **deliverables** you will need to produce.
3. Key dates and **deadlines**.
4. **Design Process Constraints** you must adhere to while working on the assignment.

INFO: Unlike constraints on the design concept(s) themselves, which are included in the Engineering Requirements for a particular design, Design Process Constraints come from the operational environment of our course, rather than directly from stakeholder needs.

EXAMPLE: For all prototyping work in the course, you must use only parts sourced from the Myhal Light Fabrication Facility (MYFab). **You are not permitted to use any parts from home, or that you purchased outside of MYFab.**

Of course, this doesn't have anything to do with the context in which your design will operate; however, it is a key constraint you should use to inform your design activities.

[1] To learn more about Process and Outcome Artifacts, please consult the Creating a Design Dossier Handout on Quercus.

2.2 Using the Rubric

A rubric is provided to help you understand how this assignment will be graded. Please consult it as you prepare your deliverables. You may also refer to the Assessment & Feedback Approach document on Quercus for more information on assessment.

2.3 Using the Template Design Dossier

For Submission #2 of the Design Dossier for the Major Design Project, we have provided a template Design Dossier which contains the following information:

- This handout;
- Template spreadsheets for required Specification and Verification+Results artifacts; and
- The required folder structure for your Design Dossier.

Your task is to design, build and test your Prototype System, and document your activities in a Design Dossier. You are expected to add both specific Outcome Artifacts as well as supporting Process Artifacts to the Design Dossier as you complete your design and fabrication work; see Section 4.2 for more information about the required artifacts you should produce.

2.4 Using your Design Dossier Submission #1 Materials

PAUSE: You should create a completely separate Design Dossier for this submission. You will **not** be adding new material to the Dossier you created in Phase 2.

The artifacts you produced during Phase 2 form the backdrop for the design and fabrication work you'll complete in Phase 3. After you've downloaded the Design Dossier Submission #2 Template, you should copy **only** these artifacts from Submission Point #1 into your new Dossier:

- Engineering Requirements Spreadsheet
- Design Concept Overview
- Prototype Concept

They should be stored in the locations indicated in Figure 1 (included later in Section 4.2).

If you need to make any updates to your framing (i.e., Engineering Requirements), Design Concept, or Prototype Concept based on new information you've learned during Phase 3, you should include new process artifacts which justify the changes you've made in the Design Dossier. There is no need to retroactively update artifacts which you created during Phase 2.

EXAMPLE: After learning new information about availability of a particular part in MYFab, your team decides to refine the Prototype Concept you proposed for Submission #1. Your team should clearly document the new information, and capture evidence which justifies why you made the changes based on that new information (as well as the revised Prototype Concept). You could add annotation to your original Prototype Concept document to explain the difference between the proposal and what you've implemented in Phase 3, and/or include additional artifact(s) which clearly describe the new Prototype Concept.

NOTE: You do not need to include detailed process information for design decisions you made during Phase 2; however, if your group made significant changes to these outcomes during Phase 3, then you should document how and why these changes came about.

3 Design Process Constraints

The Prototype System you create for this assignment should meet the Specification you will develop as a team in the early part of Phase 3. In addition to this functionality, this section outlines relevant **constraints** on the design process that your group must adhere to in your design and fabrication work.

3.1 Assignment Budget & Constraints on Sourcing Parts

The total expense budget for your Prototype System is $175.00. This includes everything you order during Phase 3, including materials/components/digital fabrication for initial iterations, testing supplies, **and** final prototype components. As for PSA, you may use components from your group members' lab kits or the MYFab Scrap Bins free of charge (*i.e.*, they do not count towards your expense budget). Unlike PSA, you are permitted to use both final sale and rental components in your Prototype System, as well as the full range of tools, equipment and materials that MYFab has to offer.

All components and materials used in all iterations of your prototype must be sourced from MYFab.

PAUSE: This constraint applies to fabrication as well as raw materials: that means that you are **not** permitted to use a personal 3D printer, laser cutter or other tools/equipment which MYFab does not offer students to borrow to fabricate any part of your prototype. For more information about this, please consult the Procurement Protocol posted on Quercus.

NOTE: When you order components as an ESC204 Project Team, you will **not** be charged out-of-pocket; the course will cover the cost. **However**, if you lose, damage or alter any rental components, you will be responsible for paying the full (final sale) cost for these components at the end of the semester!

WARNING: You may choose to put any remaining expense budget from the Major Project not spent during the term towards the cost of damaged rental components; however, if you do not have sufficient budget to cover the cost at the end of the term, you will be charged out-of-pocket for the remainder. In addition, course funds **cannot** be used to cover the cost of lost rental components.

Similar to PSA, you will submit your own CAM files to MYFab, but you will not be able to personally run the CAM equipment (3D printer or laser cutter), as the digital fabrication process

is run by MYFab staff. You should therefore ensure that your manufacturing files **and order notes** communicate your design intent and manufacturing preferences clearly.

You are permitted to use any available equipment (including 3D printers other than the Prusa Mini) when submitting DigiFab orders for the Major Project.

NOTE: Post on Piazza if you have any questions about the prototyping process and associated constraints.

3.1.1 MYFab Large Order Deadline

Sunday, March 22 will be the last day to submit large DigiFab orders to MYFab. Large orders include:

- 3D Prints with print time over 5h (using the default settings from PSA).
- Laser cuts comprising more than one full sheet of material.

Any large orders as above submitted after the deadline will not be processed. You may continue to submit small Digifab orders after this deadline; however, there is no guarantee these orders will be ready before the submission deadline for DD2.

This deadline is put in place to help MYFab manage capacity at the busy end of term, and to ensure that shared 3D Print and Laser Cutting resources remain available for teams needing to make small adjustments/updates as the submission deadline approaches.

3.2 Required Mechatronic or IoT Element(s) of the Prototype System

One of the Learning Outcomes for ESC204 states:

"By participating in ESC204, students will learn to use light fabrication methods to explore and realize design concept element(s) at high-fidelity."

In particular, it is expected that you and your team will work together to practice light fabrication skills (including 3D printing and laser cutting), work with microcontrollers to create sensor and/or actuator systems (which may communicate over a network), perform systems integration on a physical prototype, and verify its functionality. To meet these expectations, **the Prototype System you design must have significant mechatronic and/or Internet-of-Things (IoT) elements** that you will physically realize.

Recall that implementing elements of a Design Concept which are **largely or exclusively based on a policy intervention, or on the development of a machine learning/AI algorithm or other primarily software-based system (i.e., an app)** ***does not meet constraints*** for your Prototype System in ESC204. This includes mechatronic and/or IoT prototypes which have a heavy Machine Vision or Machine Learning component as a core functionality[2]. Similarly, Prototype Systems which do not include sufficient mechatronic/IoT complexity to allow all members of the team to meaningfully practice hands-on skills are not appropriate Prototype Concepts in ESC204.

[2]You may choose to explore these elements on the side, but these should not be critical path functionalities, or the core of what your prototype demonstrates.

4 Required Deliverables for Submission Point #2

This section outlines requirements for submission of this assignment. As part of the assignment, you will submit the following components:

1. A Visual Abstract (5.25% of the total 28% weight of Submission #2).
2. A Design Dossier containing both outcomes and evidence for your design, fabrication and teamwork (22.75% of the total 28% weight of Submission #2).

More details on these submissions are provided in the following sections.

4.1 Visual Abstract

Due Date: Monday, March 30 at 11:59 **AM**

Your group should submit a Visual Abstract (VA) which summarizes your Prototype System. Its purpose is to provide a high-level summary of the outcomes of your prototyping work, which may be shared with Opportunity Champions to bring the ideas you generate beyond the course.

Your Visual Abstract must include the following elements:

- ☐ A photograph or photographs of the completed Prototype System.
- ☐ A depiction of the key features of the Prototype System.
- ☐ An assessment of the function(s) and/or performance of the Prototype System relative to key elements of the specification your team developed.
- ☐ At least one representation of how the Prototype System relates to the overall Design Concept (i.e., a clear representation of which aspect(s) of the Design Concept were realized in the Prototype System, and how this realization differs from the full Design Concept).
- ☐ At least one representation of how your Design Concept fits within the context of your chosen Opportunity Space.

Your Visual Abstract must fit on **one side of one 8.5"x11" piece of paper**.

Similar to Submission Point #1, the purpose of the Visual Abstract is to provide a high-level summary of the outcomes of your project work; for Phase 3, that is your Prototype System. A reader should be able to look at the Visual Abstract and determine how your Prototype System works (i.e., its key functions), how those functions are related to the overall Design Concept, and how the Design Concept addresses stakeholder need(s). The Visual Abstract should stand alone: that is, it should be understood without reference to any other artifacts. For more detailed criteria of evaluation for the Visual Abstract, please refer to the associated rubric.

4.2 Design Dossier

Due Date: Monday, March 30 at 11:59 **AM**

Your group must also submit a Design Dossier (DD2) detailing your design and teamwork during Phase 3. The Design Dossier **must** comprise a single folder in SharePoint (or another platform), with the name `TeamNo_DD2_DesignDossier`[3].

[3] For example, if your team number is 0107E, your folder name should be `0107E_DD2_DesignDossier`.

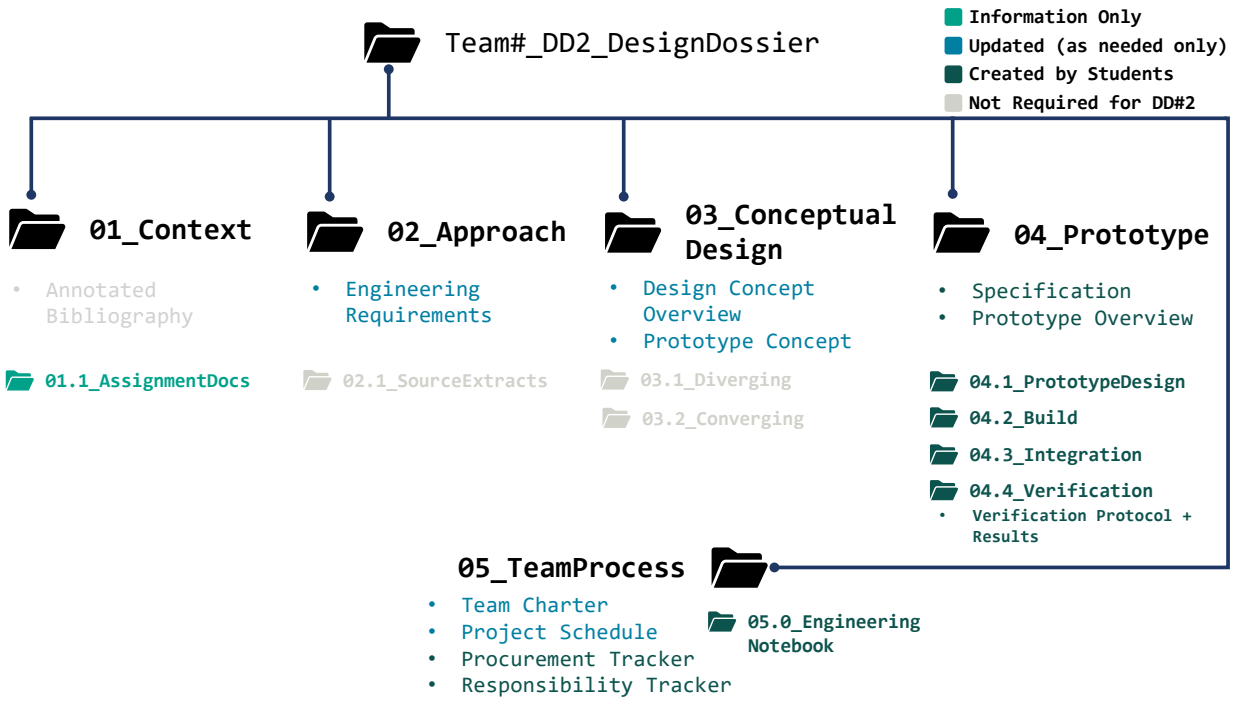

Figure 1: Overview of folder structure and required artifacts for Design Dossier Submission #1.

For Submission #2, your Design Dossier **must** contain those files and folders in Figure 1 shown in dark green ("Created by Students"), blue ("Updated (as needed only)"[4].), and light green ("Information Only"[5]). You should download the Design Dossier Template `.zip` file from Quercus to set up your Design Dossier with the appropriate structure.

The following sections provide more detailed information about key Phase 2 artifacts you should copy into the Dossier and update as appropriate for Submission #2, as well as new process, outcome and teamwork artifacts you should include from Phase 3. Required artifacts are also summarized in Tables 2–5.

Required Artifacts	**Filename** (`TeamNo_DD2_%s`)	**Folder Location**	**See Section:**
Engineering Requirements	`%s = EngineeringRequirements.xlsx`	`02_Approach`	4.2.1
Design Concept Overview	`%s = DesignConceptOverview.pdf`	`03_Conceptual Design`	4.2.1
Protoype Concept	`%s = PrototypeConcept.pdf`	`03_Conceptual Design`	4.2.1

Table 2: Summary of required Updated Artifacts reproduced and annotated (as needed) from Phase 2, and included in the Design Dossier at Submission Point #2.

[4]You should only update these documents with annotations showing changes to your framing, Design Concept, or Prototype Concept as necessary. Otherwise, copy them as-is from your Phase 2 materials (see Section 2.4).

[5]These files should be included as-is: there is no need to make any modifications.

Required Artifacts	**Filename** (`TeamNo_DD2_%s`)	**Folder Location**	**See Section:**
Specification	`%s = Specification.xlsx`	`04_Prototype`	4.2.2
Prototype Overview	`%s = PrototypeOverview.pdf`	`04_Prototype`	4.2.3
Verification Protocol & Results	`%s = VerificationProtocol+Results.xlsx`	`04.4_Verification`	4.2.4

Table 3: Summary of required Outcome Artifacts for the Design Dossier at Submission Point #2.

Required Artifacts	**Filename** (`TeamNo_DD2_%s`)	**Folder Location**	**See Section:**
Prototype Design Artifacts	Various (collection of artifacts)	`04.1_PrototypeDesign`	4.2.5
Build Artifacts	Various (collection of artifacts)	`04.2_Build`	4.2.6
Integration Artifacts	Various (collection of artifacts)	`04.3_Integration`	4.2.7
Verification Artifacts	Various (collection of artifacts)	`04.4_Verification`	4.2.8

Table 4: Summary of required Process Artifacts for the Design Dossier at Submission Point #2.

Required Artifacts	**Filename** (`TeamNo_DD2_%s`)	**Folder Location**	**See Section:**
Team Charter	`%s = TeamCharter.pdf`	`05_TeamProcess`	4.2.9
Project Schedule	`%s = ProjectSchedule.xlsx`	`05_TeamProcess`	4.2.9
Procurement Tracker	`%s = ProcurementTracker.xlsx`	`05_TeamProcess`	4.2.9
Responsibility Tracker	`%s = ResponsibilityTracker.xlsx`	`05_TeamProcess`	4.2.9
Engineering Notebook	Various (collection of artifacts)	`05.0_Engineering Notebook`	4.2.9

Table 5: Summary of required Teamwork Artifacts for the Design Dossier at Submission Point #2.

4.2.1 Updated Phase 2 Artifacts

As discussed in Section 2.4, you should upload a copy of your Requirements Spreadsheet, Design Concept Overview and Prototype Concept document to the appropriate location(s) in your Design Dossier as given in Table 2 and Figure 1. If you have not made any changes to your framing or the Design and/or Prototype Concepts you proposed at the end of Phase 2, there is no need to do anything else. These documents are provided as context to your assessor only.

If you have made significant changes to the framing (i.e., Engineering Requirements) or your Design and/or Prototype Concept(s), you should indicate these changes by annotating your original documents (for example using comments, writing notes, tracking changes etc.) to show the changes or updates you've made. If updates or changes are extensive, you should include additional artifacts in the `03_DesignConcept` folder (or a sub-folder, which you may create) which provide justification

for the changes, as well as demonstrating the new design ideas and/or framing clearly to your assessor so they can understand the context for the Prototype System you have produced.

PAUSE: **There is no need to re-write your Design Concept Overview or Prototype Concept documents, even if changes are extensive.** Instead, please use annotations or include additional (less polished) artifacts which clearly demonstrate the changes and rationale.

4.2.2 Specification for the Prototype System

You **must** submit a Specification which outlines the expected functionality and performance of your Prototype System. Your Specification should conform to the format provided in the `ESC204S_2026_DD2_Specification.xlsx` spreadsheet, located in the Design Dossier Submission #2 Template on Quercus. Specifically, your Specification must include:

- ☐ A set of numbered specification elements (i.e., individual specifications within the overall Specification spreadsheet) that your team has developed which together outline the expected functionality and performance of your Prototype System. Each numbered specification should be:
  - Clearly related to one or more top-level or subsystem objective(s) for your Design Concept, as appropriate; and
  - Written to be verifiable using the Prototype System you will actually develop.
- ☐ Justification for each specification element provided, outlining how each will demonstrate a key element of functionality or performance of the Prototype System, as well as why meeting this specification element illustrates something meaningful about the overall Design Concept.

The purpose of the Specification is to capture the intended functionality and performance of your Prototype System *before* you start to implement it, and to provide a basis for verification of the functionality and/or performance of the Prototype System *after* it has been constructed. As such, finalized elements of your Specification should also be copied into the Verification Protocol & Results spreadsheet, discussed in Section 4.2.4.

It is expected that your Specification will evolve as you design your Prototype System, so you should expect to make updates and modifications to it throughout the course; however, you should refrain from re-writing your Specification such that your Prototype System performs optimally against it after your Prototype System is complete. Your Specification should guide your design, implementation and verification work, not the other way around!

4.2.3 Prototype Overview

You **must** submit a Prototype Overview which provides a high-level summary of your design and fabrication work during Phase 3, and presents the final Prototype System (as-built). This is your opportunity to submit a more "polished" document which highlights the most important aspects of the design and fabrication work you have completed. Your Prototype Overview should include the design process you undertook to create your final Prototype System, key challenges you faced during building and integration, and an assessment of the final performance of your Prototype

System against key elements of the specification you developed. You should also discuss important design decisions you made while designing and/or fabricating your prototype, with appropriate justification.

At minimum, you should include in the Prototype Overview:

- ☐ At least one System Architecture representation of your Prototype System, distinct from that of your Design Concept.
- ☐ A photograph of your Prototype System as-built.
- ☐ A description of the detailed design of the Prototype System, with reference to artifacts elsewhere in the Dossier that can be consulted for additional information[6].
- ☐ A summary of the iterative processes your team employed to design, build, integrate and test the Prototype System, with reference to artifacts elsewhere in the Dossier which were created throughout the term.
- ☐ An assessment of the performance of the Prototype System relative to key elements of the specification you developed.
- ☐ A description of the key design decisions you made while designing and/or fabricating the Prototype System.
- ☐ Discussion of the most significant challenges you faced while creating the Prototype System.
- ☐ A description of how the Prototype System contributes to your understanding of the proposed Design Concept, including how the Design Concept will perform relative to the Need, Goals and Objectives you moved forward with from Phase 2[7].

In your Prototype Overview, you should link to the other artifacts in the `04.1_PrototypeDesign`, `04.2_Build`, `04.3_Integration`, and `04.4_Verification` folders to support claims you make about your Prototype System or design process, and to provide examples or additional detail regarding the design, build or verification processes where appropriate.

Similar to the PSA, the Prototype Overview should act as a roadmap which both summarizes the design process you undertook and key outcomes you achieved, as well as providing a guide for how a reader can navigate the additional information in the rest of your Dossier.

This document must meet the following **formatting requirements**:

- The document should be submitted in .pdf format.
- The maximum length for this document is **5 pages**, including any figures included in-text, but excluding a cover page (if desired) and list of works cited.
- The document should have at minimum 0.5” margins on all sides, and use 12pt font.

4.2.4 Verification Protocol & Results

You **must** submit a completed Verification Protocol and Results spreadsheet as part of your Design Dossier. A template you should use to set up your spreadsheet is provided in the Submission #2 Design Dossier Template on Quercus (`ESC204S_2026_DD2_Verification+Results.xlsx`). The

[6] Though elaborate figures in the text of the Integrative Summary are not required, some key figures (additional labelled photographs or screenshots from a CAD Drawing, circuit diagrams, etc.) can help to communicate your design.

[7] Or, the specific NGOs which apply to the elements of your Design Concept that you have prototyped, which may be a subset of all NGOs.

purpose of this artifact is to demonstrate whether your prototype system meets the Specification your team developed, and to house results of the verification that you and your group performed.

You should update the template spreadsheet with the elements of the Specification for your Prototype System, as well as the required method of verification for each element (i.e., Inspection, Analysis, Demonstration, or Testing). As a team, you must then:

1. Create a Verification Procedure for each specification element that will allow your team to assess how your prototype system performs relative to that element; and
2. Actually perform verification activities and record the results.

Recall that verification may be completed by Inspection (providing written and/or photographic evidence that your prototype meets the specification), through Analysis (a-priori modelling, calculations etc.), Demonstration (video or other evidence that your prototype system has met a particular specification at least once during operation), or Testing (rigorously and quantitatively demonstrating that the design meets a particular specification over multiple trials and under various operating conditions, often including off-nominal conditions and edge cases). You should be sure to select the appropriate type of verification for each element of the Specification you've developed, and indicate this in your spreadsheet. **You must choose to "Test" at least two specification elements using the Prototype System you have constructed.**

4.2.5 `04.1_PrototypeDesign`

As you design your Prototype System, you should create artifacts which represent this design process and capture these in the `04.1_PrototypeDesign` folder. You may choose to create subfolders in this space to organize your work. You should also refer to these to support claims you make in your Prototype Overview artifact. For example, you could include:

- Photograph(s)/screenshots of whiteboard(s) or other graphical collaboration tools that your group used for brainstorming at any point in the design process.
- A collaborative document where team members contributed ideas to show the evolution of your design.
- Engineering design process artifact(s) depicting trade-offs or comparisons between design ideas at any point during the design process.
- Sketches or other graphical and non-graphical representation(s) (e.g., circuit diagrams, flowcharts, state machines, pseudocode, rough CAD models) of preliminary design ideas and intended system behaviour.
- Models, calculations, or other numerical representation(s) of your system's intended performance.
- Notes from discussions with MYFab, peers, or teaching team members regarding the design of your prototype system, or selection of materials/components from the MYFab catalogue.

The idea is that artifacts in this folder show how you came up with ideas for your Prototype System, and how you determined whether your ideas were likely to meet the specification and design process constraints *before* you began construction.

4.2.6 `04.2_Build`

As you fabricate your Prototype System, you should create artifacts which represent the build processes you undertook and capture these in the `04.2_Build` folder. You may choose to create sub-folders in this space to organize your work. For example, you could include:

- The actual code which controls the behaviour of your Prototype System, and any previous versions which contributed to your understanding of its present function.
- Circuit diagram(s) you used to construct the circuit elements of your Prototype System (as-built), and any previous versions which contributed to your understanding of its present function.
- Screenshots of the CAD model(s), the actual CAD model(s) and (where relevant) drawings or sketches that you used to fabricate the structural elements of your Prototype System (as-built), and any previous versions[8].
- The CAM file(s) and screenshots of CAM model(s) submitted for DigiFab orders.
- Photos/video of each subsystem of your prototype (as-built), finalized and during fabrication.
- Notes/photos or other evidence of debugging you performed at the subsystem level to ensure that subsystems were functioning correctly prior to integration.

The idea is that artifacts in this folder show the steps you took to construct each of the Prototype System's subsystems. There should be enough information here to see exactly what components went into each of the subsystems, how they work, and any key design decisions you made as you fabricated them. You should refer to these artifacts in your Prototype Overview to support claims you make regarding the build process.

4.2.7 `04.3_Integration`

As you integrate your subsystems to create your complete Prototype System, you should create artifacts which represent the integration process and capture these in the `04.3_Integration` folder. In particular, you should include artifacts which demonstrate any challenges you overcame during integration, or any debugging you had to do to ensure that all subsystems worked together to perform their intended function(s). For example, you could include:

- Photos/video of initial versions of of your Prototype System where pieces didn't fit together, or integrated subsystems didn't achieve the desired function when taken together as a whole.
- Photos/videos or other artifacts discussing the debugging processes you undertook to correct any issues arising during integration either of the whole system or between specific subsystems.
- Artifacts capturing adjustments you could make to the design/build processes to avoid the issues you experienced with system integration next time.
- Photos/video of the fully integrated (final) Prototype System, as built.

The idea is that artifacts in this folder show how your individual subsystems came together (or failed to come together!) to form a functioning Prototype System. If your final Prototype System doesn't meet the Specification you developed, you should make sure your artifacts in this section illustrate why, and reference them to support related discussion in the Prototyping Overview.

[8] Note that if you created multiple CAD models for different design ideas **before** fabricating a final idea, these iterations may fit better in the `04.1_PrototypeDesign` folder rather than `04.2_Build`.

4.2.8 `04.4_Verification`

As you perform verification on your completed Prototype System or its subsystems, you should include additional artifacts in the `04.4_Verification` subfolder which support claims made in the Verification Protocol and Results artifact (as necessary).

4.2.9 The `05_TeamworkProcess` Folder

At the end of Phase 3, your `05_TeamworkProcess` folder **must** contain the following Process and Outcome Artifacts:

- ☐ A complete Team Charter, with revisions as appropriate.
- ☐ An expanded Project Schedule for Phase 3 of ESC204, with both intended and actual timelines.
- ☐ A Procurement Tracker artifact which shows all MYFab Orders, and accounts for all components/materials used during Phase 3 (completion only).
- ☐ A Responsibility Tracker artifact which shows who was responsible for which aspect(s) of your work during Phase 3 of the course (completion only).
- ☐ The `05.0_EngineeringNotebook` folder containing a collection of artifacts which capture a chronological account of your work during Phase 3.

The Team Charter was created during Phase 2, and should be updated to reflect your team's experiences and actual processes used throughout the course. Initial updates to your Team Charter will be facilitated during Studio. Similarly, your Project Schedule and Responsibility Tracker should be updated to reflect your team's activities throughout Phase 3; these updates will be facilitated in Studio and lecture throughout the phase.

You will also prepare a Procurement Tracker which captures all of your team's MYFab orders throughout Phase 3. This artifact is assessed for completion only; however, it must be updated regularly throughout the term to ensure your team is on-budget and help your Studio TAs understand what your team has ordered.

The remainder of this section describes the **Engineering Notebook** you should continue to create as a team as you work on your project. This will be similar to the Engineering Notebook you created as part of Phase 2; however, you should **not** include all of your old entries from Phase 2 in your Phase 3 Dossier. Instead, if you'd like to reference conversations or work that you completed during Phase 2, please include these as separate artifact(s) in the `05_TeamProcess` folder, or a subfolder.

The `05.0_EngineeringNotebook` folder should contain a set of artifacts which together comprise your team's Engineering Notebook. Though there are many interpretations of what an "Engineering Notebook" can be, in this course, your team's Engineering Notebook should be a digital, shared, chronological record of all of your team's activities during Phase 3. In particular, your Engineering Notebook should include:

- ☐ A record of each team meeting including the time/date, who was present, what was discussed (i.e., meeting minutes), and any action items.
- ☐ Logs of work done on your project, including significant work completed by the whole team, by individuals and by sub-teams. Work Log entries should include the time/date work was completed, who was present, and a summary of progress (including images, sketches, photos, links to created artifacts, questions/concerns, or other material as appropriate).

Your team may choose how to organize your Engineering Notebook to suit your particular work style, but the intention is to ensure your team captures your design process as it happens which creates a primary record of your engineering work. This means all entries must include a date and time stamp, and you **should not** remove/erase/delete information once it's been entered in the Engineering Notebook. Your notebook may be a single document or multiple documents.

Though your Engineering Notebook helps capture work as it happens, you may still place artifacts you create in other folders, and link to them in the Engineering Notebook.

EXAMPLE: Say you and a teammate work together on fabricating a structural element for a subsystem of your Prototype System in MYFab. Photos of the work you completed could be stored in the `04.2_Build` Folder (Section 4.2.6), but notes about the work session itself and a summary of what was accomplished (a Work Log) should appear in the Engineering Notebook. You could choose to include links to the photos from the Engineering Notebook to make the entry more useful to teammates who were not at the work session.

5 Submitting the Assignment

To submit the Visual Abstract and Design Dossier deliverables required for this assignment, please follow the instructions given in the sections below.

5.1 Submitting the Visual Abstract

You should upload your Visual Abstract to the Design Dossier Submission #2 – Visual Abstract assignment on Quercus before the deadline. Visual abstracts must be submitted in `.pdf` format.

5.2 Submitting the Design Dossier

As for Submission #1, you will use your team's SharePoint folder to submit your Design Dossier.

PAUSE: Please note that regardless of where it is housed, **your Design Dossier must be less than 5GB in size.** This means you should not include very high-resolution video, except in small amounts; you should also limit the number of high-resolution images you include. To meet this submission requirement, pay attention to the file size of your Dossier as you add artifacts throughout the term, and take action to reduce video file size early.

To submit, you should create a folder in your team's course SharePoint titled `TeamNo_DD2_Design Dossier`. This folder is your submitted Design Dossier: you should ensure all the artifacts you intend to be assessed are saved within this folder. Any other artifacts located elsewhere in SharePoint will not be assessed. **You must ensure that no changes are made to any artifact in this folder after 11:59AM on the due date.**

WARNING: It is your responsibility to ensure your Design Dossier is not accidentally modified after the deadline. If we cannot verify this, or your Dossier has been modified after the deadline, your submission will be considered late.

NOTE: If you have links between your documents, these will not update if you copy artifacts to a new folder. To avoid extra work to update links as the submission approaches, you may wish to work directly in your `TeamNo_DD2_DesignDossier` folder throughout Phase 3. However, it is up to you to make sure you don't accidentally edit any artifacts in your submitted Dossier as you continue to work with them after Submission Point #2 (to prepare for the Design Dossier Debrief, for example)!

If you are housing the Design Dossier in a place other than SharePoint[9] (e.g., Google Drive), you should put a link to your Design Dossier in the `TeamNo_DD2_DesignDossier` folder in SharePoint. In this case, you must also download a .zip of your complete Design Dossier, and upload this to SharePoint as well.

PAUSE: **(Only) If** you choose to link to your Design Dossier from outside of SharePoint, you **must** still submit a .zip file through SharePoint which contains identical content. Although your assessor will grade your Dossier using the linked material, if there is any dispute regarding the contents of the Dossier, or if your assessor can't access your linked files, the files in the .zip will represent your official submission.

[9] If you choose to house your Dossier outside of SharePoint, you must choose a service which is freely accessible to your assessor, without the need to sign up or create an account.